

Technical Description

Electrical Installations

Skill 18



WorldSkills International, by a resolution of the Competitions Committee and in accordance with the Constitution, the Standing Orders, and the Competition Rules, has adopted the following minimum requirements for this skill for the WorldSkills Competition.

The Technical Description consists of the following:

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1 Introduction

1.1 Name and description of the skill competition

1.1.1 The name of the skill competition is

Electrical Installations

1.1.2 Description of the associated work role(s) or occupation(s)

An electrician works on commercial, residential, agricultural, and industrial projects. There is a direct relationship between the nature and quality of the product required and the payment made by the customer. Therefore, the electrician has a continuing responsibility to work professionally in order to meet the requirements of the customer and thus maintain and grow the business.

Electrical installation is closely associated with other parts of the construction industry and with the many products that support it, normally for commercial purposes.

The electrician works internally or in teams, in the homes of customers and on small and major projects. They will plan and design, select and install, commission, test, report, maintain, fault find, and repair systems to a high standard. Work organization and self-management, communication, and interpersonal skills, problem solving, flexibility and a deep body of knowledge are the universal attributes of the outstanding electrician.

With a constantly developing technology an electrician will face new challenges where new systems will be needed and new working methods have to be used.

Whether the electrician is working alone or in a team the individual takes on a high level of personal responsibility and autonomy. From working to provide a safe and reliable electrical installation and maintenance service, in accordance with relevant standards, through to diagnosing malfunctions, programming and commissioning home and building automation systems and creating documentation, concentration, precision, accuracy, and attention to detail at every step in the process matters. Mistakes are largely irreversible, costly, and potentially life threatening.

With the international mobility of people, the electrician faces rapidly expanding opportunities and challenges. For the talented electrician there are many commercial and international opportunities; however, these carry with them the need to understand and work with diverse cultures and trends. The diversity of skills associated with electrical installations is therefore likely to expand.

An electrician also has many career opportunities including advancing to leading or managerial positions.

1.1.3 Number of Competitors per team

Electrical Installations is a single Competitor skill competition.

1.1.4 Age limit of Competitors

The Competitors must not be older than 22 years in the year of the Competition.

1.2 The relevance and significance of this document

This document contains information about the standards required to compete in this skill competition, and the assessment principles, methods, and procedures that govern the competition.

Every Expert and Competitor must know and understand this Technical Description.

In the event of any conflict within the different languages of the Technical Descriptions, the English version takes precedence.

1.3 Associated documents

Since this Technical Description contains only skill-specific information it must be used in association with the following:

- WSI – Code of Ethics and Conduct
- WSI – Competition Rules
- WSI – WorldSkills Occupational Standards framework
- WSI – WorldSkills Assessment Strategy
- WSI online resources as indicated in this document
- WorldSkills Health, Safety, and Environment Policy and Regulations
- WorldSkills Standards and Assessment Guide (skill-specific)

2 The WorldSkills Occupational Standards (WSOS)

2.1 General notes on the WSOS

The WSOS specifies the knowledge, understanding, skills, and capabilities that underpin international best practice in technical and vocational performance. These are both specific to an occupational role and also transversal. Together they should reflect a shared global understanding of what the associated work role(s) or occupation(s) represent for industry and business (www.worldskills.org/WSOS).

The skill competition is intended to reflect international best practice as described by the WSOS, to the extent that it can. The Standard is therefore a guide to the required training and preparation for the skill competition.

In the skill competition the assessment of knowledge and understanding will take place through the assessment of performance. There will only be separate tests of knowledge and understanding where there is an overwhelming reason for these.

The Standard is divided into distinct sections with headings and reference numbers added.

Each section is assigned a percentage of the total marks to indicate its relative importance within the Standards. This is often referred to as the “weighting”. The sum of all the percentage marks is 100. The weightings determine the distribution of marks within the Marking Scheme.

Through the Test Project, the Marking Scheme will assess only those skills and capabilities that are set out in the WorldSkills Occupational Standards. They will reflect the Standards as comprehensively as possible within the constraints of the skill competition.

The Marking Scheme will follow the allocation of marks within the Standards to the extent practically possible. A variation of up to five percent is allowed, if this does not distort the weightings assigned by the Standards.

2.2 WorldSkills Occupational Standards

Section		Relative importance (%)
1	Work organization and management	5
	The individual needs to know and understand: • Health and safety legislation, obligations, and documentation • The principles of working safely with electricity • The situations when personal protective equipment (PPE) must be used • The purposes, uses, care, maintenance, and storage of all tools and equipment together with their safety implications • The purposes, uses, care, and storage of materials • The importance of keeping a tidy work area • Sustainability measures applying to the use of ‘green’ materials and recycling	

Section		Relative importance (%)
	• The ways in which working practices can minimize wastage and help to manage costs whilst maintaining quality • The principles of workflow and measurement • The significance of planning, accuracy, checking, and attention to detail in all working practices • The impact of new technology.	
	The individual shall be able to: • Develop and follow Health, Safety, and Environment standards, rules, and regulations • Diligently follow electrical safety procedures • Identify and use the appropriate personal protective equipment (PPE) including safety footwear, ear, and eye protection • Select, use, clean, maintain, and store all tools and equipment safely • Select, use, and store all materials safely • Identify and take care of expensive fixtures/fittings • Plan the work area to maximize efficiency and maintain the discipline of regular tidying • Measure accurately • Manage time effectively • Work efficiently and check progress and outcomes regularly • Establish and consistently maintain high quality standards and working processes.	
2	Communication and interpersonal skills	5
	The individual needs to know and understand: • The significance of establishing and maintaining customer confidence and trust • The importance of maintaining and keeping knowledge base up-to-date • The roles and requirements of related trades • The value of building and maintaining productive working relationships • Techniques of effective teamwork • The importance of swiftly resolving misunderstandings and conflicting demands.	
	The individual shall be able to: • Interpret customer requirements and manage customer expectations positively • Provide advice and guidance on products/solutions e.g. technological advancements • Visualize and translate customer wishes making recommendations which meet/improve their design and budgetary requirements • Question customers closely/deeply to fully understand requirements	

Section		Relative importance (%)
	• Provide clear instructions • Introduce related trades to support customer requirements • Produce written reports for customers and the organization • Produce cost and time estimates for customers • Recognize and adapt to the changing needs of related trades • Work effectively as a member of a team.	
3	Problem solving, innovation, and creativity	5
	The individual needs to know and understand: • The common types of problem which can occur within the work process • Diagnostic approaches to problem solving • Trends and developments in the industry including new technology, standards, and working methods e.g. “smart house” and energy saving measures • Potential problems with procurement and alternative solutions.	
	The individual shall be able to: • Check work regularly to minimize problems at a later stage • Identify problems originating from the work of a related trade e.g. heating pump, ventilation system, etc. • Challenge incorrect information to prevent problems • Recognize and analyze problems swiftly and follow a self-managed process for resolving • Recognize opportunities to contribute ideas to improve solutions and overall levels of customer satisfaction • Show willingness to try new methods and embrace change e.g. ready-made components • Recommend to customers alternative solutions for better, smarter and more cost efficient and sustainable installations.	
4	Planning and design	10
	The individual needs to know and understand: • Different types of standards, drawings, installation descriptions, and manuals • A range of materials and installation techniques to be used in different environments.	
	The individual shall be able to: • Read, interpret, and revise drawings and documentation including layout and circuit drawings • Follow written instructions • Plan installation work using drawings and documentation provided.	

Section		Relative importance (%)
5	Installation	35
	The individual needs to know and understand: • Ducting and wiring systems for commercial, domestic, residential agricultural, and industrial use and when and where to use a specific ducting and/or wiring system • The range of electrical switchboards used for commercial, domestic, residential, agricultural, and industrial uses and when and where to use a specific switchboard system • Types of electric lighting and heating systems for commercial, domestic residential, and industrial use • Control devices and socket outlets used for commercial, domestic, residential, agricultural, and industrial uses, including smart building technologies • Structured cabling systems including: computer network cabling, fire/burglar alarm (conventional and addressable), evacuation control (audio and optical), control and monitoring, access control ("stand-alone" and 'network supervised'), closed circuit television (cameras, lenses and attachment components), recorders and monitors • Energy production systems such as solar- and wind-power • Systems for charging of Electrical Vehicles.	
	The individual shall be able to: • Select and install equipment and wire ways according to drawings and documentation provided • Install ducting and cabling systems on different surfaces according to manufacturers' instructions and current industrial standards • Select and install single and double insulated cables inside ducts, conduits, and flexible conduits • Install and securely fix double insulated cables onto cable ladders, cable trays and different surfaces according to manufacturers' instructions and current industrial standards • Install metal and plastic ducting (trunking); accurately measure and cut ducts at specified lengths/angles; assemble without distortion to joints and to specified tolerances • Assemble different termination adaptors, including glands onto ducts and attach ducts, of different types, securely onto surfaces • Install metal and plastic conduits/flexible conduits and accessories and attach securely onto surfaces, maintaining even radius bends, without distortion to conduits if manually bent. • Use the correct termination adaptors used for entry of conduits into boxes, boards, and ducts • Install and securely attach different types of cable ladders and cable trays to surfaces • Install electrical switchboards onto surfaces in a secure way and assemble switchboard apparatus in switchboards according to layout drawings/instructions to include main switches, RCDs,	

Section		Relative importance (%)
	MCBs, fuses, controlling equipment such as relays and timers and home and building automation devices • Terminate and install wiring inside switchboards according to circuit drawings • Connect equipment as per instructions provided to include structured cabling systems according to manufacturers' instructions and current industrial standards and regulations • Install systems such as electrical car chargers, solar panels, energy management systems and other related systems for a sustainable future.	
6	Testing, reporting, and commissioning	25
	The individual needs to know and understand: • Industrial regulations and standards applicable to different types of installations • Verification standards, methods, and reports to be used to record verification results • Types of measuring instruments • Tools and software used for parameterization, programming, and commissioning • The correct operation of the electrical installation in accordance with the planned specification and customer requirements • The importance of delivering correct and proper documentation "as built" after finalized installations for future reference and maintenance purposes.	
	The individual shall be able to: • Test installations before energizing to ensure personal and electrical safety to include insulation resistance and earth continuity tests, correct polarity, and complete visual inspections • Test installations when energized by checking complete function on all equipment installed to ensure correct operation of new installations as per instructions, for example, correct voltage, phase rotation and correct functioning of protection devices • Set-up equipment to include: selecting and using the appropriate software for programming programmable relays, bus-systems; creating necessary settings on devices such as timers and overload relays; programming programmable relays: downloading and importing applications required and programming bus-systems • Set installations to fully functioning and ensure customers can operate • Provide data for updating drawings and other related documentation after finalized installation work.	

Section		Relative importance (%)
7	Maintenance, fault finding, and repair	15
	The individual needs to know and understand: • Different types of installations for specific environments • Different generations of installations • The purpose of specific installations • The customers' needs for various functions.	
	The individual shall be able to: • Adapt to changing circumstances • Troubleshoot electrical installations and identify faults including short and open circuits, incorrect polarity, insulation resistance and earth continuity faults, incorrect settings on equipment, and incorrect program on programmable devices • Diagnose electrical installations and identify problems including bad connections, incorrect wiring, high loop impedance, and equipment failure • Verify that existing electrical installations still meet current standards • Use, test and calibrate measuring equipment including insulation resistance, continuity, and installation testers, multi, clamp and network cable testers • Repair and replace faulty components in electrical installations • Rewire and or repair faulty installations • Recycle replaced equipment in a correct and sustainable way.	
	Total	100

3 The Assessment Strategy and Specification

3.1 General guidance

Assessment is governed by the WorldSkills Assessment Strategy. The Strategy establishes the principles and techniques to which WorldSkills assessment and marking must conform.

Expert assessment practice lies at the heart of the WorldSkills Competition. For this reason, it is the subject of continuing professional development and scrutiny. The growth of expertise in assessment will inform the future use and direction of the main assessment instruments used by the WorldSkills Competition: the Marking Scheme, Test Project, and Competition Information System (CIS).

Assessment at the WorldSkills Competition falls into two broad types: Measurement and Judgement. For both types of assessment, the use of explicit benchmarks against which to assess each Aspect is essential to guarantee quality.

The Marking Scheme must follow the weightings within the Standards. The Test Project is the assessment vehicle for the skill competition, and therefore also follows the Standards. The CIS enables the timely and accurate recording of marks; its capacity for scrutiny, support, and feedback is continuously expanding.

The Marking Scheme, in outline, will lead the process of Test Project design. After this, the Marking Scheme and Test Project will be designed, developed, and verified through an iterative process, to ensure that both together optimize their relationship with the Standards and the Assessment Strategy. They will be agreed by the Experts and submitted to WSI for approval together, to demonstrate their quality and conformity with the Standards.

Prior to submission for approval to WSI, the Marking Scheme and Test Project will liaise with the WSI Skill Advisors for quality assurance and to benefit from the capabilities of the CIS.

4 Assessment Design and Practice

4.1 General guidance

This section describes the role and place of the Marking Scheme, how the Experts will assess Competitors' work as demonstrated through the Test Project, and the procedures and requirements for marking.

The Marking Scheme is the pivotal instrument of the WorldSkills Competition, in that it ties assessment to the standard that represents each skill competition, which itself represents a global occupation. It is designed to allocate marks for each assessed aspect of performance in accordance with the weightings in the Standards.

By reflecting the weightings in the Standards, the Marking Scheme establishes the parameters for the design of the Test Project. Depending on the nature of the skill competition and its assessment needs, it may initially be appropriate to develop the Marking Scheme in more detail as a guide for Test Project design. Alternatively, initial Test Project design can be based on the outline Marking Scheme. From this point onwards the Marking Scheme and Test Project should be developed together.

Section 2.1 above indicates the extent to which the Marking Scheme and Test Project may diverge from the weightings given in the Standards, if there is no practicable alternative.

For integrity and fairness, the Marking Scheme and Test Project are increasingly designed and developed by one or more Independent Test Project Designer(s) with relevant expertise. In these instances, the Marking Scheme and Test Project are unseen by Experts until immediately before the start of the skill competition, or competition module. Where the detailed and final Marking Scheme and Test Project are designed by Experts, they must be approved by the whole Expert group prior to submission for independent validation and quality assurance. Please see the Competition Rules for further details.

Experts and Independent Test Project Designers are required to submit their Marking Schemes and Test Projects for review, verification, and validation well in advance of completion. They are also expected to work with their Skill Advisor, reviewers, and verifiers, throughout the design and development process, for quality assurance and in order to take full advantage of the CIS's features.

In all cases a draft Marking Scheme must be entered into the CIS at least eight weeks prior to the Competition. Skill Advisors actively facilitate this process.

4.2 Assessment Criteria

The main headings of the Marking Scheme are the Assessment Criteria. These headings are derived before, or in conjunction with, the Test Project. In some skill competitions the Assessment Criteria may be similar to the section headings in the Standards; in others they may be different. There will normally be between five and nine Assessment Criteria. Whether or not the headings match, the Marking Scheme as a whole must reflect the weightings in the Standards.

Assessment Criteria are created by the person or people developing the Marking Scheme, who are free to define the Criteria that they consider most suited to the assessment and marking of the Test Project. Each Assessment Criterion is defined by a letter (A-I). **The Assessment Criteria, the allocation of marks, and the assessment methods, should not be set out within this Technical Description. This is because the Criteria, allocation of marks, and assessment**

methods all depend on the nature of the Marking Scheme and Test Project, which is decided after this Technical Description is published.

The Mark Summary Form generated by the CIS will comprise a list of the Assessment Criteria and Sub Criteria.

The marks allocated to each Criterion will be calculated by the CIS. These will be the cumulative sum of marks given to each Aspect within that Assessment Criterion.

4.3 Sub Criteria

Each Assessment Criterion is divided into one or more Sub Criteria. Each Sub Criterion becomes the heading for a WorldSkills marking form. Each marking form (Sub Criterion) contains Aspects to be assessed and marked by Measurement or Judgement, or both Measurement and Judgement.

Each marking form (Sub Criterion) specifies both the day on which it will be marked, and the identity of the marking team.

4.4 Aspects

Each Aspect defines, in detail, a single item to be assessed and marked, together with the marks, and detailed descriptors or instructions as a guide to marking. Each Aspect is assessed either by Measurement or by Judgement.

The marking form lists, in detail, every Aspect to be marked together with the mark allocated to it. The sum of the marks allocated to each Aspect must fall within the range of marks specified for that section of the Standards. This will be displayed in the Mark Allocation Table of the CIS, in the following format, when the Marking Scheme is reviewed from C-8 weeks. (Section 4.1 refers.)

TOTAL MARKS	STANDARDS SPECIFICATION SECTION	CRITERIA								TOTAL MARKS PER SECTION	WSSS MARKS PER SECTION	VARIANCE	
			A	B	C	D	E	F	G	H			
		1	5.00								5.00	5.00	0.00
		2		2.00					7.50		9.50	10.00	0.50
		3								11.00	11.00	10.00	1.00
		4			5.00						5.00	5.00	0.00
		5				10.00	10.00	10.00			30.00	30.00	0.00
		6		8.00	5.00				2.50	9.00	24.50	25.00	0.50
		7			10.00				5.00		15.00	15.00	0.00
			5.00	10.00	20.00	10.00	10.00	10.00	15.00	20.00	100.00	100.00	2.00

4.5 Assessment and marking

There is to be one marking team for each Sub Criterion, whether it is assessed and marked by Judgement, Measurement, or both. The same marking team must assess and mark all Competitors. Where this is impracticable (for example where an action must be done by every Competitor simultaneously, and must be observed doing so), a second tier of assessment and marking will be put in place, with the approval of the Competitions Committee Management Team. The marking teams must be organized to ensure that there is no compatriot marking in any circumstances. (Section 4.6 refers.)

4.6 Assessment and marking using Judgement

Judgement uses a scale of 0-3. To apply the scale with rigour and consistency, Judgement must be conducted using:

- benchmarks (criteria) for detailed guidance for each Aspect (in words, images, artefacts, or separate guidance notes). This is documented in the Standards and Assessment Guide.
- the 0-3 scale to indicate:
 - 0: performance below industry standard
 - 1: performance meets industry standard
 - 2: performance meets and, in specific respects, exceeds industry standard
 - 3: performance wholly exceeds industry standard and is judged as excellent

Three Experts will judge each Aspect, normally simultaneously, and record their scores. A fourth Expert coordinates and supervises the scoring, and checks their validity. They also act as a judge when required to prevent compatriot marking.

4.7 Assessment and marking using Measurement

Normally three Experts will be used to assess each Aspect, with a fourth Expert supervising. In some circumstances the team may organize itself as two pairs, for dual marking. Unless otherwise stated, only the maximum mark or zero will be awarded. Where they are used, the benchmarks for awarding partial marks will be clearly defined within the Aspect. To avoid errors in calculation or transmission, the CIS provides a large number of automated calculation options, the use of which is mandated.

4.8 The use of Measurement and Judgement

Decisions regarding the choice of criteria and assessment methods will be made during the design of the competition through the Marking Scheme and Test Project.

4.9 Skill assessment strategy and procedures

WorldSkills is committed to continuous improvement including reviewing past limitations and building on good practice. The following skill assessment strategy and procedures for this skill competition take this into account and explain how the marking process will be managed.

The Test Project assessment is based on the following criteria:

A - Personal safety during work and electrical safety on the completed installations on all Modules;

B - Testing, reporting, and commissioning from every module is assessed as described in the instructions for the various modules;

C - Circuit design is assessed on the functionality of the circuit and the cable selection. Both for safety and cost efficiency;

D - Measurements and level/plumb is assessed comparing drawings with the actual installations;

E - Installation of equipment wire-ways may be checked as but not limited to;

F - Wiring and terminations may focus on but not limited to;

G - Installation testing is assessed as faults found or not found;

H - Programming is assessed as functions completed or not completed.

5 The Test Project

5.1 General notes

Sections 3 and 4 govern the development of the Test Project. These notes are supplementary.

Whether it is a single entity, or a series of stand-alone or connected modules, the Test Project will enable the assessment of the applied knowledge, skills, and behaviours set out in each section of the WSOS.

The purpose of the Test Project is to provide full, balanced, and authentic opportunities for assessment and marking across the Standards, in conjunction with the Marking Scheme. The relationship between the Test Project, Marking Scheme, and Standards will be a key indicator of quality, as will be its relationship with actual work performance.

The Test Project will not cover areas outside the Standards or affect the balance of marks within the Standards other than in the circumstances indicated by Section 2. This Technical Description will note any issues that affect the Test Project's capacity to support the full range of assessment relative to the Standards. Section 2.1 refers.

The Test Project will enable knowledge and understanding to be assessed solely through their applications within practical work. The Test Project will not assess knowledge of WorldSkills rules and regulations.

Most Test Projects and Marking Schemes are now designed and developed independently of the Experts. They are designed and developed either by the Skill Competition Manager, or an Independent Test Project Designer, normally from C-12 months. They are subject to independent review, verification, and validation. (Section 4.1 refers.)

The information provided below will be subject to what is known at the time of completing this Technical Description, and the requirement for confidentiality.

Please refer to the current version of the Competition Rules for further details.

5.2 Format/structure of the Test Project

The Test Project is a series of two (2) standalone modules.

Module	Description
Module 1	Domestic and commercial installation, testing and reporting, and programming
Module 2	Surprise task (project changes)

5.3 Test Project design requirements

Test Projects should reflect the purposes, structures, processes, and outcomes of the occupational role they are based on. They should aim to be a small-scale version of that role. Before focusing on practicalities, SMTs should show how the Test Project design will provide full, balanced, and authentic opportunities for assessment and marking across the Standards, as set out in Section 5.1.

General requirements

- Test Project to be designed by industry partners to reflect current and upcoming technologies;
- All technical terms and descriptions used in the Test Project must be in accordance with internationally recognized standards;
- Cable and conduit measurements are to the centre of the cable/conduit;
- Trunking and equipment measurements are to the centre of the trunking/equipment;
- All dimensions must be from specific reference lines (datum/centre lines);
- Different cable types must be used. For example:
 - Plastic sheath cable;
 - Flexible cable;
 - Steel wire armour;
- Different cable support systems must be used. For example:
 - Metal conduit;
 - PVC conduit;
 - Metal cable tray;
 - PVC trunking.

The Test Project will consist of the following modules

Module 1 – Domestic and commercial installation and programming, testing, and reporting

- This module will run over C1, C2, C3, and C4;
- The Competition Organizer must supply all the materials for this module;
- Measurement marks are measured progressively each day as indicated by the Mark Summary Form;
- Module one is installed on three walls and the ceiling of the Competitors cubicle;
- Module one will include home and building automation and a small smart relay device; as well as conventional wiring systems
- Competitors are to be provided with descriptions, other necessary documentations and associated product files (product data base);
- The Infrastructure List must state the languages that the software, software version, and databases are available in;
- Programming exercise is to be completed in front of the general public;
- Final function tasks decided upon at the Competition;
- Programming is limited to the control of motors, sockets, lighting, HVAC and blinds;
- Programming functions to be listed in a table along with images of devices;
- Competitors are forbidden to have electronic storage devices in their workstation during this module.
- Test commissioning is part of this module.
- Fault finding,

Module 2 – Surprise task (project change)

- Time allocated to be included in the overall time of the Competition
- Module 2 will be a modification to Module 1 in one of the following ways
 - Change of functionality;
 - Addition to installation;
 - Modification of installation;
 - Material replacement.

General instructions for all modules

The Test Projects must reflect the IEC standards and not one specific continent. Once all the Test Projects have been completed, they must reflect aspects of electrical installations throughout the world.

Test specifications:

- Earth continuity resistance – The maximum resistance between the main incoming earth terminal and any point on the installation required to be earthed may not be more than 0.5 Ω ;
- Insulation resistance – The minimum resistance between any current carrying conductors and any other conductors to earth may not be less than one M Ω , tested at a voltage of 500 V DC with an insulation resistance tester;
- Polarity of socket outlets as per Host Country standard.

Requirements for Module 1 – Domestic and commercial installation and commissioning

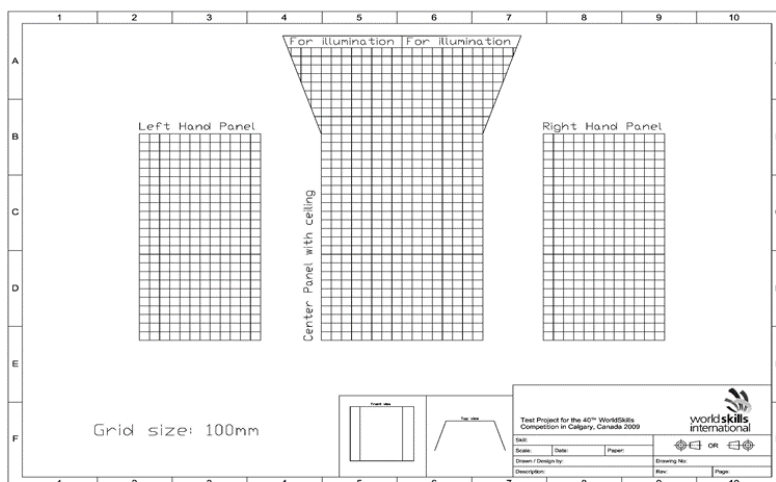
- This module must include lighting circuits, power outlet circuits;
 - This module must include a design task for the Competitor;
 - Installation of distribution boards and protection equipment must be included;
 - Installation of programmable devices must be included;
 - Installation of home and building automation devices must be included;
 - This module may include fixed appliance circuits, Structured Cabling Systems, environment control or access equipment;
 - Inspection and testing is carried out and the test results documented before commissioning;
 - Function testing may be carried out with the installation live at standard mains voltage of the Host Country
-
- The test circuit designs can include the following circuits:
 - A lighting circuit;
 - A socket outlet circuit;
 - A power circuit (such as a heater or a cooker);
 - A control circuit (such as a pump control);
 - Types of faults that may also be :
 - Incorrect timer settings;
 - Incorrect overload settings;
 - Short circuit faults;
 - Open circuit faults;
 - High resistance joints;
 - Interconnection;

Competition Organizer requirements

- Ensure a power supply of 230/400 V AC at each workstation;
- Discrimination to be provided between each Competitors bay;
- Ensure that necessary power supplies are available for testing;
- Will build some installation samples for module one in the Competitors area of the competition floor.

Panel layout

The layout of the work cubicle shown below is just for reference purposes.



5.4 Test Project coordination and development

The Test Project MUST be submitted using the templates provided by WorldSkills International (www.worldskills.org/expertcentre). Use the Word template for text documents and DWG template for drawings.

5.4.1 Test Project coordination (preparation for Competition)

Coordination of the Test Project/modules will be undertaken by the Skill Competition Manager.

5.4.2 Who develops the Test Project/modules

The Test Project/modules are developed by an Independent Test Project Designer (ITPD) in collaboration with the Skill Competition Manager.

5.4.3 When is the Test Project developed

The Test Project/modules are developed according to the following timeline:

Time	Action
Fifteen (15) months prior to the Competition	The ITPD is identified and a Confidentiality Agreement between WSI and the ITPD is organized.
No later than two (2) months prior to the Competition	The Test Project documents are sent to the WorldSkills International Skills Competitions Administration Manager.
At the Competition on C-2	The secret items on the Infrastructure List are presented to Experts and Competitors.
At the Competition on C1	The Test Project is presented to Experts and Competitors.

5.5 Test Project initial review and verification

The purpose of a Test Project is to create a challenge for Competitors which authentically represents working life for an outstanding practitioner in an identified occupation. By doing this, the Test Project will apply the Marking Scheme and fully represent the WSOS. In this way it is unique in its context, purpose, activities, and expectations.

To support Test Project design and development, a rigorous quality assurance and design process is in place (Competition Rules sections 10.6-10.7 refer.) Once approved by WorldSkills, the Independent Test Project Designer (ITPD) is expected to identify one or more independent expert(s), and trusted individuals initially to review the Independent Test Project Designer's ideas and plans, and subsequently to verify the Test Project, prior to validation.

A Skill Advisor will ensure and coordinate this arrangement, to guarantee the timeliness and thoroughness of both initial review, and verification, based on the risk analysis that underpins Section 10.7 of the Competition Rules.

5.6 Test Project validation

The Skill Competition Manager coordinates the validation of the Test Project/modules and will ensure that it can be completed within the material, equipment, knowledge, and time constraints of Competitors.

5.7 Test Project circulation

The Test Project/modules are not circulated prior to the Competition. The Test Project/modules are presented to Experts and Competitors on C1.

5.8 Test Project change

Due to the Test Project being developed by an Independent Test Project Designer (ITPD), there is no change required to be made to the Test Project/modules at the Competition. Exceptions are amendments to technical errors in the Test Project documents and according to infrastructure limitations.

5.9 Test Project translation and use of Interpreters

The Test Project documentation and the Mark Summary Form may be translated into the language of the Competitors' choice, however, it is the Member's responsibility to translate the Test Project documents, either by a compatriot Interpreter or by the Expert. Refer to the Skill Management Plan for time allocation for translation.

For more information refer to **Competition Rule 10.12**.

6 Skill management and communication

6.1 Discussion Forum

Prior to the Competition, all discussion, communication, collaboration, and decision making regarding the skill competition must take place on the WorldSkills skill-specific Discussion Forum. (<http://forums.worldskills.org>). Skill related decisions and communication are only valid if they take place on the WorldSkills Discussion Forum. The Chief Expert (or an Expert Lead appointed by the Skill Management Team) will be the moderator for this Discussion Forum. Refer to the Competition Rules for the timeline of communication and competition development requirements.

6.2 Competitor information

All information for registered Competitors is available from the Competitor Centre (www.worldskills.org/competitorcentre).

This information includes:

- Competition Rules
- Technical Descriptions
- Mark Summary Form (where applicable)
- Test Projects (where applicable)
- Infrastructure List
- WorldSkills Health, Safety, and Environment Policy and Regulations
- Other Competition-related information

6.3 Test Projects and Marking Schemes

Circulated Test Projects will be available from www.worldskills.org/testprojects and the Competitor Centre (www.worldskills.org/competitorcentre).

6.4 Day-to-day management

The day-to-day management of the skill competition during the Competition is defined in the Skill Management Plan that is created by the Skill Management Team. The Skill Management Team comprises the Skill Competition Manager, Chief Expert, and the Expert Leads. The Skill Management Plan is progressively developed in the six (6) months prior to the Competition and finalized at the Competition. The Skill Management Plan can be viewed in the Expert Centre (www.worldskills.org/expertcentre).

6.5 General best practice procedures

General best practice procedures clearly delineate the difference between what is a best practice procedure and skill-specific rules (section 9). General best practice procedures are those where Experts and Competitors CANNOT be held accountable as a breach to the Competition Rules or skill-specific rules which would have a penalty applied as part of the Issue and Dispute Resolution procedure including the Code of Ethics and Conduct Penalty System. In some cases, general best practice procedures for Competitors may be reflected in the Marking Scheme.

Topic/task	Best practice procedure
Use of technology	• Note: spy software may be used on computers.
Equipment failure	• If there is clear evidence that Competitors have caused damage to equipment themselves, they will not be given a substitute and will not be given any additional time.

7 Skill-specific safety requirements

7.1 Personal Protective Equipment

Refer to WorldSkills Safety Policy and Regulations for Host country or region regulations.

Task	Safety glasses with side protection	Leather/heat resistant gloves	Cut-resistant gloves	Tight fitting work clothes (long trousers)	Insulated safety shoes with protective cap (Competitors only)	Sturdy shoes with closed toe and no heel	Hearing protection
General PPE for safe areas				√		√	
Drilling (Does not include screwing)	√		√	√	√		√
Sawing	√		√	√	√		√
Working with heat gun	√	√		√	√		√
General work				√	√		

8 Materials and equipment

8.1 Infrastructure List

The Infrastructure List details all equipment, materials, and facilities provided by the Competition Organizer.

The Infrastructure List is available at www.worldskills.org/infrastructure.

The Infrastructure List specifies the items and quantities requested by the Skill Management Team for the next Competition. The Competition Organizer will progressively update the Infrastructure List specifying the actual quantity, type, brand, and model of the items. Note that in some cases details of specific materials and/or manufacturer specifications may remain secret and will not be released prior to the Competition. These items may include those for fault finding modules or modules not circulated.

At each Competition, the Skill Management Team must review and update the Infrastructure List in preparation for the next Competition. The Skill Competition Manager must advise the Director of Skills Competitions of any increases in space and/or equipment.

At each Competition, the Technical Observer must audit the Infrastructure List that was used at that Competition for the upcoming WorldSkills Competition.

The Infrastructure List does not include items that Competitors and/or Experts are required to bring and items that Competitors are not allowed to bring – they are specified below.

8.2 Material or manufacturer specifications

Specific material and/or manufacturer specifications required to allow the Competitor to complete the Test Project will be supplied by the Competition Organizer and are available from www.worldskills.org/infrastructure located in the Expert Centre. However, note that in some cases details of specific materials and/or manufacturer specifications may remain secret and will not be released prior to the Competition. These items may include those for fault finding modules or modules not circulated.

If specific material or manufacturer specifications are required to allow the Competitor to complete the Test Project, it is provided along with the Test Project, the Workshop Manager must arrange a demonstration on site during Familiarization Day.

The materials chosen for modules that are to be built by Competitors, except where the materials are to be supplied by the Competitor, should be of a type available from a number of manufacturers and readily obtainable from suppliers in the Competition Organizer.

8.3 Competitors toolbox

Competitors are not allowed to send a toolbox to the Competition. All tools are provided by the Competition Organizer as per Infrastructure List available one year before the Competition.

8.4 Materials, equipment, and tools supplied by Competitors

It is not applicable for Competitors to bring materials, equipment, and tools to the Competition.

However, Competitors are allowed to bring stationery, toolbelts, calculators, clocks, and their workwear in the morning of C-2 on Familiarization Day. It is recommended that these tools be brought in the luggage of the Competitor or purchased locally.

Furthermore, Competitors are required to supply their own Personal Protective Equipment as specified in section 7 skill-specific safety requirements.

8.5 Materials, equipment, and tools supplied by Experts

Experts are required to supply their own Personal Protective Equipment as specified in section 7 skill-specific safety requirements.

Experts are responsible that Interpreters bring their own PPE.

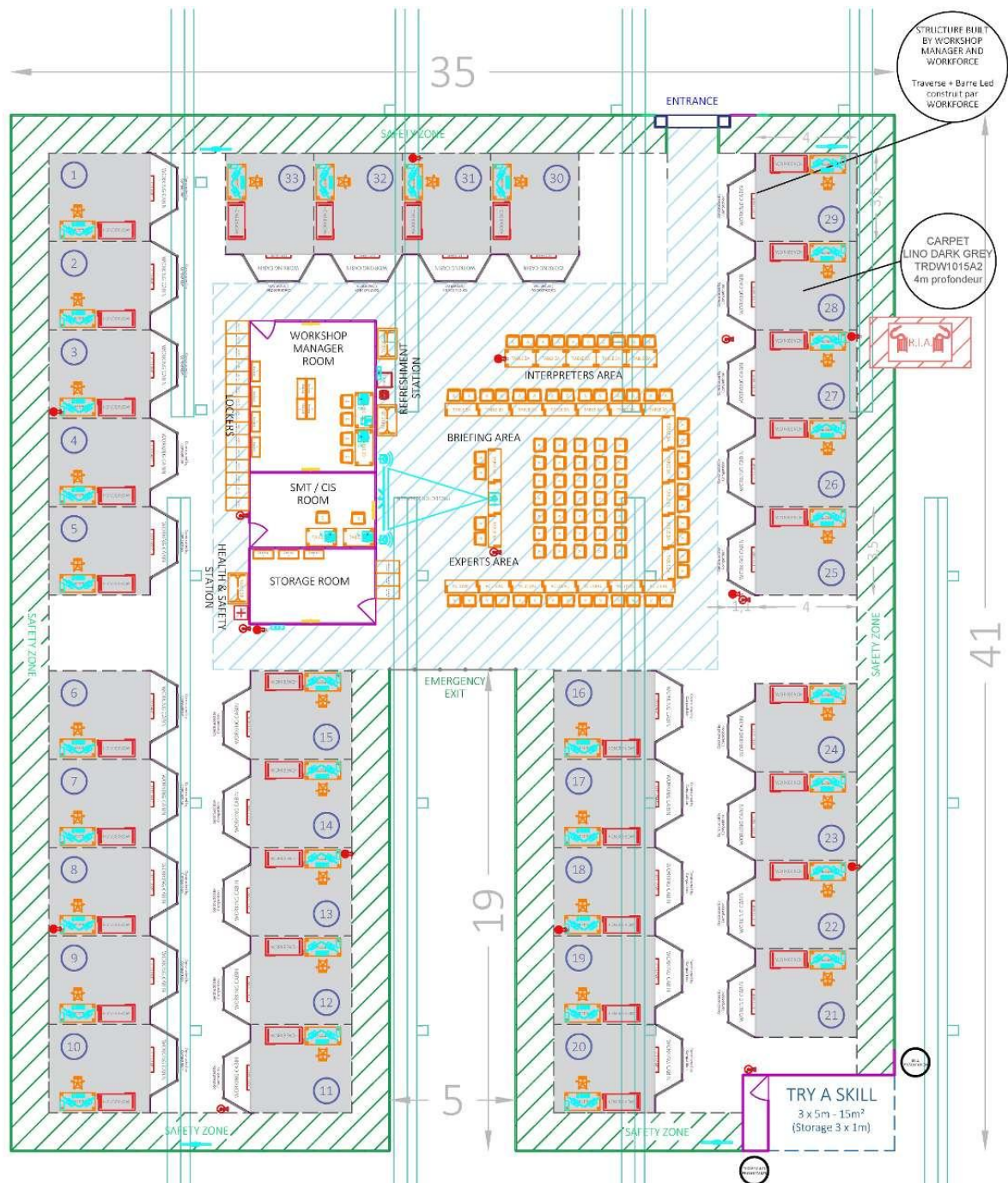
8.6 Materials and equipment prohibited in the skill area

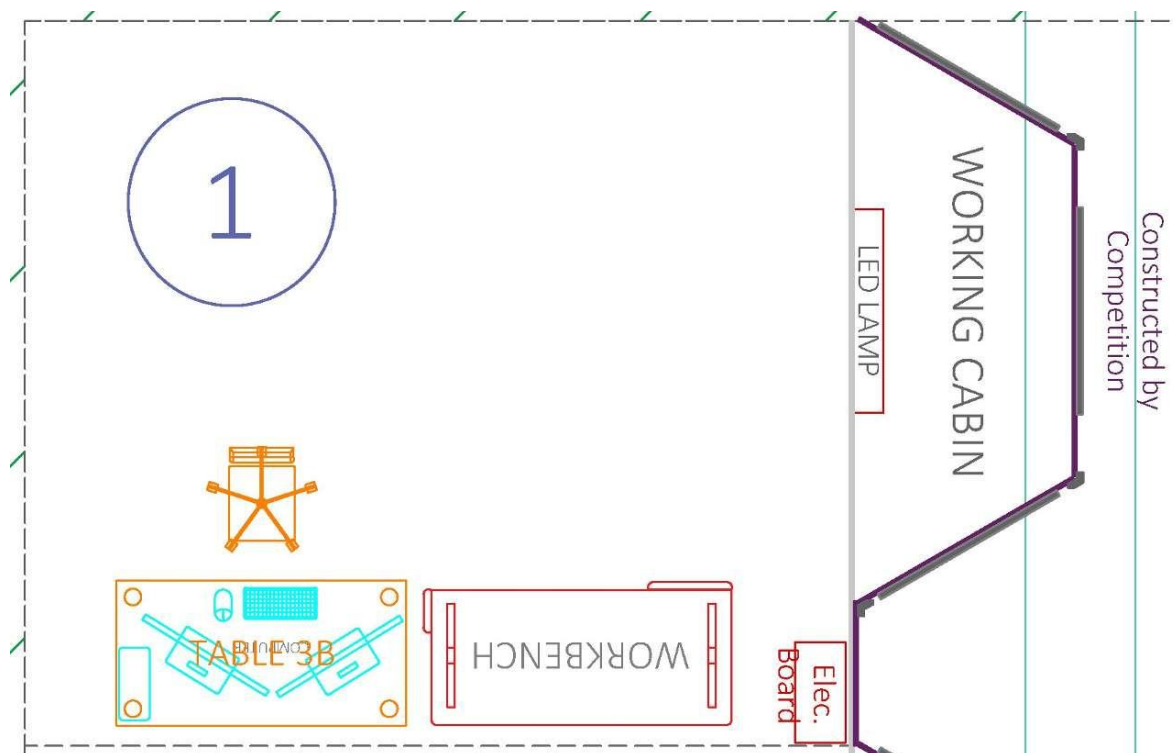
Competitors and Experts are prohibited to bring any materials or equipment not listed in section 8.3 and section 8.4.

8.7 Proposed workshop and workstation layouts

Workshop layouts from previous competitions are available at www.worldskills.org/sitelayout.

Example workshop layout





9 Skill-specific rules

9.1 General notes

Skill-specific rules cannot contradict or take priority over the Competition Rules. They do provide specific details and clarity in areas that may vary from skill competition to skill competition. This includes but is not limited to personal IT equipment, data storage devices, Internet access, procedures and workflow, and documentation management and distribution. Breaches of these rules will be solved according to the Issue and Dispute Resolution procedure including the Code of Ethics and Conduct Penalty System.

9.2 Skill-specific rules

Topic/task	Skill-specific rules
Use of technology – USB, memory sticks	• Competitors are only allowed to use memory sticks provided by the Competition Organizer. No other memory sticks are to be inserted into the Competitor computers. • Memory sticks or any other portable memory devices cannot be taken outside the workshop. • Memory sticks or other portable memory devices are to be submitted to the Chief Expert at the end of each day for safe keeping and must not be taken out of the workshop. • Only Competitors and Technical Support Team are allowed to work with the computers in Competitors workstation unless special permission is given by the SMT. No Expert can be given such permission to their compatriot Competitor.
Use of technology – personal laptops, tablets, and mobile phones	• Competitors are not allowed to bring personal laptops, tablets, or mobile phones into the workshop on C-2, C1 to C4. If Competitors do bring these items into the workshop they must place them in their locker. They can use them at break times and take at the end of each day. Experts, Chief Expert, Skill Competition Manager and Interpreters are exempt from this rule. • Competitors, Chief Experts, Skill Competition Manager, Experts, and Interpreters are allowed to use personal photo and video taking devices in the workshop, however no photos can be taken of details of the Test Project documents and material or any marking forms. • Skill Competition Manager and Workshop Manager are able to take photos to support technical matters. • Competitors may bring a wired mouse and wired keyboard in a language of their choice into the workshop. NOTE: A wired QWERTY keyboard and mouse will be provided.

Topic/task	Skill-specific rules
Drawings, recording information	• Competitors may make drawings, document instructions, or take notes whilst in the workshop however they cannot be taken out of the workshop at any time. • Chief Expert, Experts, and Interpreters may make notes to support the marking process.
Supervision of Competitors	• Competitors must be supervised at all time during their work. Experts on supervising duties must ensure another Expert replaces them if they have to leave. • Experts are not allowed to supervise their compatriot Competitor. • Experts and Interpreters are only allowed to enter a workstation if approved by the Chief Expert. The only exception to this rule is if it's necessary to stop a Competitor for Health, Safety, and Environment reasons.

10 Expert knowledge and experience

10.1 Requirements

Experts appointed for this skill competition must have the following knowledge and experience for the appropriate occupation or work role as documented in **section 1.1.2**.

To provide leadership, technical oversight, assessment integrity, and coaching to a WorldSkills Electrical Installations Competitor, ensuring performance at global standards.

Minimum qualifications (≥ 3 – 5 years):

- Qualified electrician certification or equivalent;
- Advanced technical qualification;
- Certified in training and assessment;
- Vocational teaching or coaching in TVET institutions, technical centres, or apprenticeship programs.

Industry experience (≥ 5 years):

- Hands-on experience across domestic, commercial, agricultural, and industrial installations;
- Proficiency in wiring systems, switchboards, structured cabling systems, EV charging, and renewable technologies;
- Familiarity with international standards like IEC, BS 7671, AS/NZS codes.

Specialist competency areas

Criterion	Performance Levels
Technical competence	Installation and testing skills in mock or live competitions, including smart systems.
Safety and organization	Strict adherence to safety protocols and work area discipline during assessment.
Interpersonal and communication	Clarity in instructions, customer communication, teamwork, and report writing.
Design and planning skills	Effective reading and interpretation of schematics, and planning solutions.
Fault presence and problem-solving	Ability to anticipate, identify, and resolve installation and system errors.
Standards and regulation knowledge	Up-to-date application of international codes (IEC, BS 7671, AS/NZS) and norms.
Mentoring and Competitor support	Skilful coaching and constructive feedback that supports Competitor growth.

11 Visitor and media engagement

11.1 Engagement methods

Following is a list of possible ways to maximize visitor and media engagement:

- Try-a-Skill - an area next to the workshop, supervised by local apprentices, where young people can try some of the things an electrician work with on a daily basis;
- Display screens;
- Test Project descriptions;
- Drawings and Test Projects/parts of Test Projects may be displayed next to the Try-a-Skill.
- Enhanced understanding of Competitor activity;
- Competitor profiles may be displayed on screens close to the competition site. Useful information is:
 - Name;
 - Age;
 - Country of origin;
 - Type of education;
 - Type of actual career;
 - Information about the Competitors' choice of apprenticeship;
- Career opportunities;
- Information may consist of:
 - Brochures;
 - Flyers;
 - Informers (young apprentices);
- Daily reporting of competition status if all Competitors are working on the same module on the same day.

12 Sustainability

12.1 Sustainable practices

This skill competition will focus on the sustainable practices below:

- Recycling;
- Spare materials are to be offered to local schools to be used in education for training purposes;
- Use of “green” materials;
- During designing of Test Projects and liaising with the current Workshop Manager, the uses of “green” materials are to be considered. Materials that meet the following requirements are to be used as far as possible:
 - Halogen free;
 - Recyclable;
 - Free from toxic substances.
- Use of completed Test Projects after Competition;
- Materials that can be reused, are to be offered to local schools to be used in education for training purpose. Materials that cannot be reused are to be sorted as per Host Country regulations or, if meeting higher requirements, WorldSkills regulations;
- The Independent Test Project Designer must carefully consider sustainability as a key issue;
- All paperwork prepared at the previous competition must be electronically copied by the new Chief Expert.

13 References for industry consultation

13.1 General notes

WorldSkills is committed to ensuring that the WorldSkills Occupational Standards fully reflect the dynamism of internationally recognized best practice in industry and business. To do this WorldSkills approaches a number of organizations across the world that can offer feedback on the draft Description of the Associated Role and WorldSkills Occupational Standards on a two-yearly cycle.

In parallel to this, WSI consults three international occupational classifications and databases:

- ISCO-08: (<http://www.ilo.org/public/english/bureau/stat/isco/isco08/>)
- ESCO: (<https://ec.europa.eu/esco/portal/home>)
- O*NET OnLine (www.onetonline.org/)

13.2 References

This WSOS appears most closely to relate to Electrician:

<https://www.onetonline.org/link/summary/47-2111.00>

or Domestic Electrician:

<http://data.europa.eu/esco/occupation/5dbb9cf0-b226-402c-a295-2f42ef05ff8b>

These links can also be used to explore adjacent occupations.

ILO 7411.

The following table indicates which organizations were approached and provided valuable feedback for the Description of the Associated Role and WorldSkills Occupational Standards in place for WorldSkills Shanghai 2026.

Organization	Contact name
INCE.Swiss	Norbert Ivan Büchel, Head of Vocational training

14 Appendix

14.1 Appendix information

Not applicable.